FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – XI
PAPER –2
TEST DATE: 23-08-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 –06, 20 – 25, 39 – 44)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A ***(07 –10, 26 – 29, 45 – 48)*****:** This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A solid sphere and a solid cube of same material, same volume are heated to the same temperature and allowed to cool in the same surroundings.

(A) The ratio of the initial rate of cooling of the sphere and the cube is $\left(\frac{\pi}{3}\right)^{1/3}$

(B) The ratio of the initial rate of cooling of the sphere and the cube is $\left(\frac{\pi}{6}\right)^{1/3}$

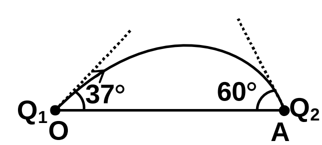
(C) The ratio of the initial power radiated by the sphere and the cube is $\left(\frac{3}{4\pi}\right)^{2/3}$

(D) The ratio of the initial power radiated by the sphere and the cube is $\left(\frac{\pi}{6}\right)^{1/3}$

2. Hydrogen atom in its ground state is excited by means of monochromatic radiations. The resulting emission spectrum has six different lines. These radiations are incident on a metal plate. It is observed that only two of them are responsible for photoelectric effect. If the ratio of maximum kinetic energies of photoelectrons in the two cases is $\frac{11}{10}$, then

(A) The work function of the metal is 5.48 eV.
(B) The work function of the metal is 2.49 eV.
(C) The ratio of maximum wavelenght to the minimum wavelength in the emission spectrum is 15 : 4.
(D) The ratio of maximum wavelenght to the minimum wavelength in the emission spectrum is 135 : 7.

3. Two point charge Q_1 and Q_2 are placed along x-axis at origin O (0, 0) and position A(a, 0). An electric field line emerges from q, at an angle 37° to the x-axis and terminates the charge q_2 at angle 60° with the x-axis as shown. Then (Take cos 37° = 4/5)



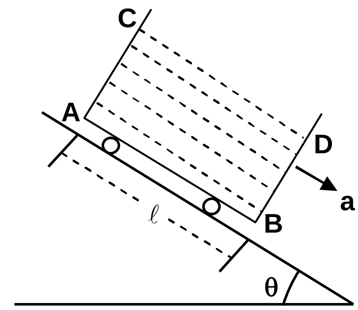
(A) Net electric field due to the two point charges is zero at position $\left(\frac{\sqrt{5}a}{\sqrt{5}-\sqrt{2}}, 0\right)$

(B) Net electric potential is zero on axis at positions $\left(\frac{5a}{3}, 0\right)$ and $\left(\frac{5a}{7}, 0\right)$

(C) The locus of points in xy plane where net potential is zero is a circle with centre at $\left(\frac{25a}{21}, 0\right)$ and radius $\frac{10a}{21}$

(D) $\frac{Q_1}{Q_2} = -\frac{2}{5}$

4. A container filled partially with a liquid of density ρ is sliding on an inclined plane of inclination θ with acceleration $a = \frac{g}{n}$, (n > 1) where n is a constant. The length of the base of the container is AB = ℓ. Then



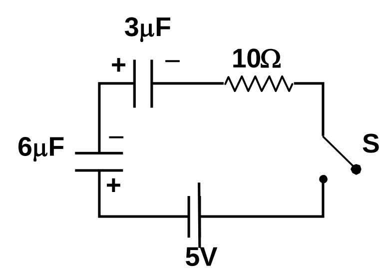
(A) The angle made by the free surface CD of the liquid with horizontal = $\tan^{-1}\left(\frac{\cos\theta}{n+\sin\theta}\right)$

(B) The angle made by the free surface CD of the liquid with the inclined plane = $\left|\theta - \tan^{-1}\left(\frac{\cos\theta}{n-\sin\theta}\right)\right|$

(C) The pressure difference of liquid at points A and B, $P_B - P_A = \rho g\ell\left(\sin\theta - \frac{1}{n}\right)$

(D) The pressure difference of liquid at points A and B, $P_B - P_A = \rho g\ell\left(\frac{1}{n} - \sin\theta\right)$

5. In the circuit shown in the figure, initially the switch S is open and the capacitors of capacitance 3μF and 6 μF are charged to a potential difference of 2V and 3V respectively with their polarity as shown. The battery of emf 5V is ideal. Then



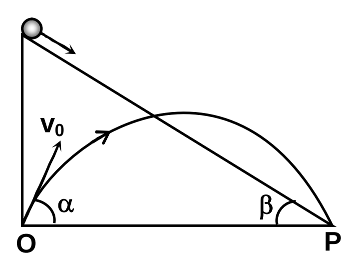
(A) The current in the circuit just after the switch is closed, is 2.5 A

(B) The charge flow through the cell from its negative to positive terminal after the switch is closed is +20μC.

(C) The magnitude of final potential difference on the 3μF and 6μF after the switch is closed are $\frac{14}{3}$V and $\frac{1}{3}$V respectively.

(D) The total heat produced in the circuit after the switch S is closed, is 100 μJ

6. Two point objects one sliding down from rest on a frictionless inclined plane of inclination β with the horizontal the other being projected in air from the point O at an angle α with horizontal start their motion at the same instant. Initially both point objects are along the same vertical line. Both get to the point P at the same time and the same speed. Then choose the correct option(s).



(A) $2\sin\alpha = \frac{1}{\sin\beta}$

(B) $2\cos\alpha = \cos\beta$

(C) $\sin\alpha = \frac{2}{\sin\beta}$

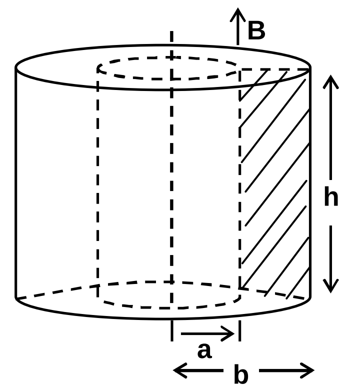
(D) $\cos\alpha = 2\cos\beta$

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 07 and 08

A ring of a rectangular cross section (as shown in the figure) is made of a material whose resistivity is ρ. A uniform magnetic field B exists parallel the axis of the ring in the entire region and increases with time as B = kt where k is a constant. Find the current induced in the ring.



7. The current induced in the ring is

(A) $\dfrac{kh(b^2-a^2)}{\rho}$

(B) $\dfrac{kh(b^2-a^2)}{4\rho}$

(C) $\dfrac{kh(b^2-a^2)}{2\rho}$

(D) $\dfrac{2kh(b^2-a^2)}{\rho}$

8. Now the time varying magnetic field is switch off and potential difference V is applied between the inner and outer curved surface of the ring. The current in the ring will be

(A) $\dfrac{\pi hV}{\rho \ell n\left(\frac{b}{a}\right)}$

(B) $\dfrac{2\pi h\rho}{V \ell n\left(\frac{b}{a}\right)}$

(C) $\dfrac{2\rho hV}{\pi \ell n\left(\frac{b}{a}\right)}$

(D) $\dfrac{2\pi hV}{\rho \ell n\left(\frac{b}{a}\right)}$

Paragraph for Question Nos. 09 and 10

A projectile is projected from the origin O on the level ground at an angle of 30° with an initial speed of 40 m/s. At one point during its trajectory the projectile explodes into two pieces. The two pieces reach the ground at same moment. One of them hits the ground at exactly where it was projected, with a speed of $20\sqrt{2}$ m/s. At what height the collision occur? Neglect air resistance. (Given g = 10 m/s^2). Take x-axis as horizontal direction and y-axis as vertically upward direction. The motion takes place in xy vertical plane.

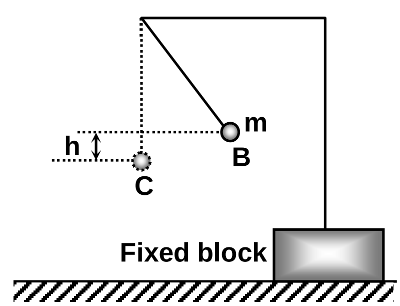
9. The angle made by the velocity of the piece with horizontal when it hits the origin 'O'.
   (A) 45°
   (B) 30°
   (C) 60°
   (D) 37°

10. The x-coordinate of the position where the projectile exploeds is
   (A) 31.75 m
   (B) 25.30 m
   (C) 50.76 m
   (D) 41.75 m

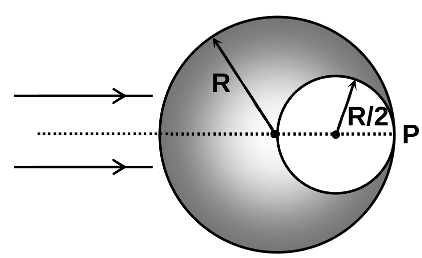
Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. A small positively charged ball B of mass m is suspended by an insulating thread of negligible mass. Another positively charged small ball C is moved slowly from a large distance until it is in the original position of the first ball. As a result the ball B rises by h. How much work (in joule) has been done by the agent moving the charge C? (Given m = 1kg, g = 10 m/s^2 and h = 10 cm)



12. A transparent sphere of radius R has a cavity of radius R/2 as shown in the figure. A parallel beam of light falling on the left surface focuses at point P. Consider the incident rays to be paraxial. If the refractive index of the material of the sphere is μ. Find the value of 100μ. (Take $\sqrt{5} = 2.24$)



13. A pipe P open at one end and closed at the other end is filled with hydrogen gas at temperature 27°C. An another pipe Q is filled with a mixture of gases consisting of 1 mole of helium and 1 moles of oxygen gases at temperature 77°C. Then length of the pipe Q is $\sqrt{5}$ times the length of pipe A and it is open at both ends. The air column in pipe P is vibrating in first overtone and the air column in pipe Q is vibrating in second overtone. If the ratio of the frequencies of vibrations of air column in pipes P and Q is n : 1, find the value of n.

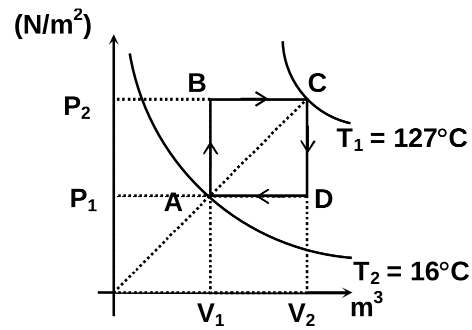
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

Between two isotherms, a cyclic process ABCDA is performed with one mole of ideal mono-atomic gas. If the work done by the gas in one complete cycle is X Joule and temperature of gas at state B is T_B = Y kelvin. (Take gas constant R = 8.31 $Jmol^{-1}K^{-1}$)



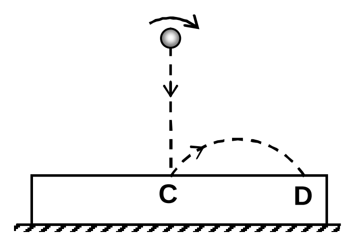
14. The value of X is………..

15. The value of Y is………….

Question Stem for Question Nos. 16 and 17

Question Stem

A solid sphere of mass m = 1 kg and radius R = 0.1 m is spun about a horizontal axis and then dropped without an initial speed onto a stationary plank of mass M = 2kg from a height h = 5 m. It hits the plank exactly at the centre C. The sphere keep sliding on the plank throughout the duration of collision. The coefficient of kinetic friction between the sphere and the plank is μ = 0.1. The magnitude of vertical component of velocity of centre of the ball just before and just after collision is equal. After the sphere rebounds it again strike the plank at point D. The distance CD is X meter and change in angular velocity of sphere due to the first collision with the plank is Y rad/s.



Frictionless surface

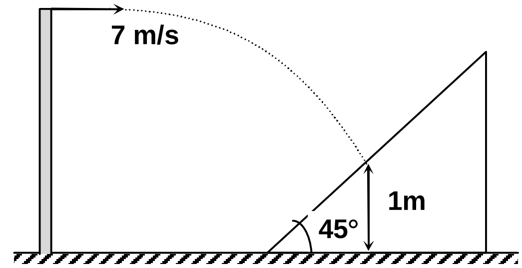
16. The value of X is………..

17. The value of Y is………….

Question Stem for Question Nos. 18 and 19

Question Stem

A ball of mass 1 kg is projected horizontally with velocity 7 m/s from a tower of height 3.5 m. It collides elastically with a wedge of mass 3 kg and inclination 45° kept on a horizontal frictionless surface. Ball colides the wedge at a height of 1 m above the ground. If the velocity of the wedge just after collision is X m/s and magnitude of vertical component of impulse exerted by the ball on the wedge during the collision is Y N-s. (Take g = 9.8 m/s^2)



18. The value of X is……………..

19. The value of Y is………….

Chemistry

PART – II

Section – A (Maximum Marks: 24)

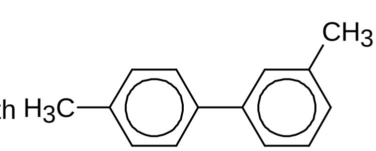
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

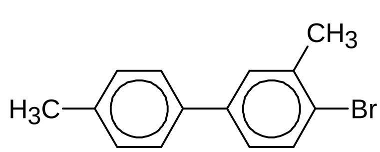
20. Which of the following is/are correct option for 2s orbital?
    (A) It has 1 radial node
    (B) $\psi_R = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_o}\right)^{3/2}\frac{Zr}{a_o}e^{-Zr/2a_o}$
    (C) At radial node, $\frac{Zr}{a_o} = 2$
    (D) Angular function of $2s = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$

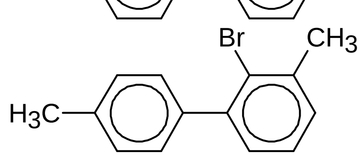
21. Select the correct option/s
    (A) for N_2 LUMO is π^*2p_x and π^*2p_y
    (B) for F_2 HOMO is π^*2p_x and π^*2p_y
    (C) for O_2^{--} HOMO is σ^*2p_z
    (D) for C_2 LUMO is $\sigma 2p_z$

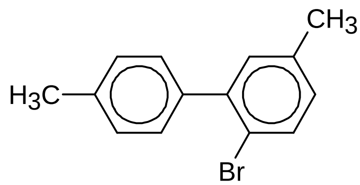
22. Which of the following is/are Transuranium element?
    (A) Th
    (B) Pa
    (C) Np
    (D) Bk

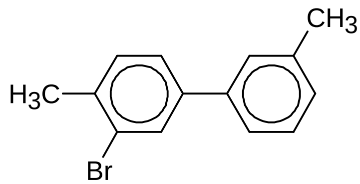
23. NaCl is doped with 10^{-3} mole % $FeCl_3$. Select the correct option for the given information
    (A) The number of unoccupied octahedral voids per mole of NaCl is 6.02×10^{18}
    (B) The number of unoccupied tetrahedral voids per mol of NaCl is 12.04×10^{18}
    (C) The cation vacancy for per mole of NaCl is 12.04×10^{18}
    (D) The number of unoccupied octahedral voids for 100 moles of NaCl is 12.04×10^{20}

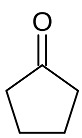
24. Possible product/s obtained when Br_2 / Fe is treated with  is/are

    (A)


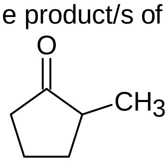
    (B)


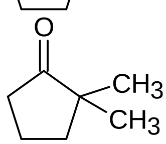
(C)


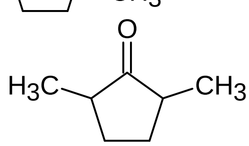
(D)


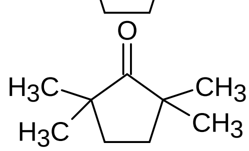
25.  $\xrightarrow{OH^-/CH_3Br}$ Product / s

Possible product/s of above reaction is/are

(A)


(B)


(C)


(D)


Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 26 and 27

Kohlrausch law states that limiting molar conductance of an electrolyte can be represented as the sum of the individual contributions of the anion and the cation of the electrolyte. Thus, if $\lambda^o_{Na^+}$ and $\lambda^o_{Cl^-}$ are limiting molar conductivity of sodium and chloride ions respectively, then the limiting molar conductivity of sodium chloride is given by the equation

$\Lambda^o_{m(NaCl)} = \lambda^o_{Na^+} + \lambda^o_{Cl^-}$

In general, if an electrolyte on dissociation gives ν_+ cation and ν_- anions then its limiting molar conductivity is given by

$\Lambda^o_m = \nu_+\lambda^o_+ + \nu_-\lambda^o_-$

26. Select the correct increasing order of limiting molar conductivity

(A) $\lambda^{o}_{OH^-} < \lambda^{o}_{H^+} < \lambda^{o}_{Na^+}$

(B) $\lambda^{o}_{OH^-} < \lambda^{o}_{Na^+} < \lambda^{o}_{H^+}$

(C) $\lambda^{o}_{Na^+} < \lambda^{o}_{H^+} < \lambda^{o}_{OH^-}$

(D) $\lambda^{o}_{Na^+} < \lambda^{o}_{OH^-} < \lambda^{o}_{H^+}$

27. The conductivity of 0.001028 mol L^{-1} of HA is $4.95 \times 10^{-5}\ \text{Scm}^{-1}$. Calculate its dissociation constant if Λ^{o}_{m} for HA is 390.5 $\text{S cm}^2\ \text{mol}^{-1}$

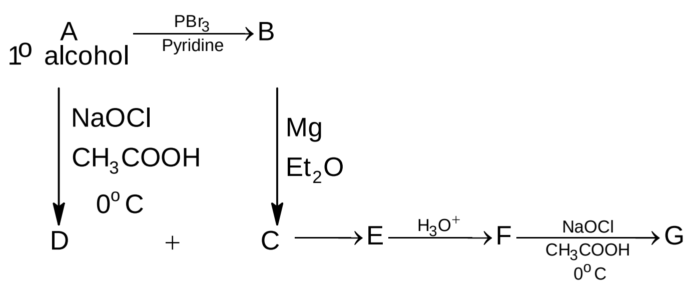
(A) 1.56×10^{-5}

(B) 1.78×10^{-5}

(C) 2×10^{-4}

(D) 1.56×10^{-4}

Paragraph for Question Nos. 28 and 29



28. Select the correct statement

(A) G can give Tollen's test

(B) D can't give Tollen's test

(C) F is secondary alcohol

(D) C is tetrahydrofuran

29. $D \xrightarrow[\text{HCl}]{^{-}C\equiv N(\text{excess})} H \xrightarrow[\Delta]{\text{HCl, } H_2O} I$

(A) I is $R-\overset{\overset{OH}{|}}{\underset{\underset{R}{|}}{C}}-COOH$

(B) I is $R-\overset{\overset{OH}{|}}{CH}-COOH$

(C) I is $R-\overset{\overset{OH}{|}}{CH}-CH_2Cl$

(D) I is $R-\overset{\overset{OH}{|}}{\underset{\underset{R}{|}}{C}}-CH_2Cl$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. For the parallel chain reaction the rate of formation of B is 50% of rate of decay of A and rate of formation of C is 40% of rate of decay of A, what will be the partial half-life of A in hrs. while it is converting into D (Given that A parallelly gives B, C and D and overall half-life of A is 150 hrs.

31. A metal block of 270 gram having total heat capacity $566\ JK^{-1}$ at $100^{o}C$ is placed in a lake at $10^{0}C$. Calculate entropy change of lake $\left(\ell n1.317 = 0.276\right)$

32. 36 ml of an aqueous solution of KCl was found to require 20 ml of 1 M $AgNO_3$ solution. When titrated using a suitable indicator. Depression in freezing point of KCl solution with 80% ionization will be: $\left(K_f = 2\ Kkg\ mol^{-1} \text{ and molarity} = \text{molality}\right)$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

The specific rotation of enantiomerically pure (+)-2-butanol is +13.5^{o}. If

$x = \dfrac{ee}{0.005}$ and $y = \dfrac{\text{observed rotation}}{0.0003}$ then for a mixture containing 6g of (+)-2-butanol and 4g of (-)-2-butanol calculate the following

33. Calculate the value of x ________

34. The value of y is_________

Question Stem for Question Nos. 35 and 36

Question Stem

p-methyl benzaldehyde, p-dimethyl aminobenzaldehyde and p-nitrobenzaldehyde show self cannizzaro reaction among all, one p-substituted benzaldehyde is least reactive as compared to benzaldehyde at $100^{o}C$.

If x = degree of unsaturation of oxidized form of least reactive p-substituted benzaldehyde $\times 10^{3}$ and

y = molecular wt. of most reactive p-substituted benzaldehyde $\times \dfrac{x}{200}$, then

[Given: C = 12, O = 16, N = 14, H = 1]

35. The value of x is_________

36. The value of y is_________

Question Stem for Question Nos. 37 and 38

Question Stem

An aqueous solution of 10 moles of KIO_3 is treated with an excess of KI solution. This is done in acidic condition and liberates I_2, which is consumed in 20 litres of sodium thiosulphate solution. Here,

$$x = \frac{\text{strength of hypo solution} \times 100}{2}$$

and y = milliequivalents of $I_2 \times n_f$ of I_2 obtained from reaction between KI and $KIO_3 \times \frac{1}{40}$.

37. The value of x is__________

38. The value of y is__________

Mathematics PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. Let $\sum_{k=1}^{\infty} \operatorname{cosec}^{-1}\left(\frac{\sqrt{k^2+4k+3}}{\sqrt{k+2}-\sqrt{k}}\right) = \alpha$, then (where [.] denotes the greatest integer function)

(A) $\frac{d}{dx}\left(\sin^{-1}\frac{2x}{1+x^2}\right)$ at $x = \tan\left(\frac{3\pi}{4}-\alpha\right)$ is $-\frac{2}{3}$

(B) $\int_0^{\frac{\left(\alpha+\tan^{-1}\sqrt{2}\right)}{2}} [\tan x]dx = \frac{\pi}{2} - \tan^{-1}2$

(C) $\lim_{x\to\tan\left(\frac{3\pi}{4}-\alpha\right)} \left(1+\sin\left(x-\sqrt{2}\right)\right)^{\frac{x}{\tan\left(x-\sqrt{2}\right)}} = e^{\sqrt{2}}$

(D) $\int_0^{\tan^2\left(\frac{3\pi}{4}-\alpha\right)} \left[\tan^{-1}x\right]dx = 2-\tan 1$

40. Let $f(x) = \sin^{-1}\left(\frac{9-x^2}{9+x^2}\right) + \cos^{-1}\left(\frac{6x}{9+x^2}\right)$, then

(A) total number of local maximum of f(x) in (–6, 6) are 3

(B) $\int_{-3}^{3} f(x)dx = 3\pi + 6\ln 2$

(C) total number of local minimum of f(x) in (–6, 6) are 3

(D) number of points where f(x) is non differentiable in (–6, 6) are 3

41. Let $A = \begin{bmatrix} \sqrt{3} & 2 \\ 2 & -\sqrt{3} \end{bmatrix}$ and B be a non singular matrix of order such that $A^{-1} = A^T - A.\mathrm{Adj}(2B^{-1})$ (A^{-1} and A^T represent inverse and transpose of matrix A respectively), then

(A) $\mathrm{Det}\left(\mathrm{Adj}\left(2B^{-1}\right)\right) = \frac{49}{9}$

(B) $\mathrm{Det}\left(\mathrm{Adj}\left(2B^{-1}\right)\right) = \frac{36}{49}$

(C) $\mathrm{Det}\, B = \frac{49}{9}$

(D) $\mathrm{Det}\left(B^{-1}\right) = \frac{9}{40}$

42. Let l, m, n be three real numbers satisfying $[lmn]\begin{bmatrix} 3 & +1 & 6 \\ -1 & 2 & +5 \\ 4 & -3 & -5 \end{bmatrix} = [3\ \ -2\ \ -3]$ ..... S . If the point P(l, m, n) with reference to S lies on a plane (π) containing the lines $L_1 : \frac{x-1}{1} = \frac{y-2}{-2} = \frac{z-3}{-1}$ and $L_2 : \frac{x-1}{4} = \frac{y-2}{-3} = \frac{z-3}{0}$, then which of the following option(s) are correct?

(A) distance of plane (π) from the point $\left(0,-1,-\sqrt{2}\right)$ is 1

(B) line $L:\frac{x+\frac{2}{3}}{4}=\frac{y+\frac{1}{2}}{-3}=\frac{z-0}{0}$ completely lies in plane (π)

(C) $5\ell+2m-n=2$

(D) Image of (l, m, n) about L_2 is (8, 3, 8)

43. Let f(x) be a double differentiable function such that $|f''(x)| \le 10$ in [1, 6] and f(x) assumes its maximum strictly within the interval (1, 6), then $|f'(1)| + |f'(6)|$ can assume the value(s)
(A) 20
(B) 30
(C) 40
(D) 60

44. An ellipse having eccentricity $\frac{3}{4}$ passes through the point P(5, 4). If the nearer focus is (1, 1) and equation of tangent at P on the ellipse 4x + 3y – 32 = 0, then
(A) length of latus rectum of ellipse is 17.5
(B) length of chord through (1, 1) parallel to the tangent at P is 17.5
(C) length of perpendicular from the focus other than (1, 1) on the tangent at P is 35
(D) locus of point of intersection of perpendicular tangents drawn to ellipse is $(x + 11)^2 + (y + 8)^2 = 575$

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 45 and 46

Consider $f(x)=\left|\sin\left(\left(3a-1\right)\frac{\pi}{4}\right)x\right|-\left|\cos\left(\frac{b\pi}{4}\right)x\right|$. If a, b are the number obtained on throwing a fair die two times, then

45. The probability that f(1) is a non negative integer is
(A) $\frac{1}{2}$
(B) $\frac{5}{12}$
(C) $\frac{7}{12}$
(D) $\frac{1}{3}$

46. The probability that $f\left(\frac{4}{3}\right)$ is an irrational number is
(A) $\frac{1}{2}$
(B) $\frac{5}{12}$

(C) $\frac{7}{12}$
(D) $\frac{2}{3}$

Paragraph for Question Nos. 47 and 48

Let a_1, a_2, a_3, ….. a_n be n terms of an increasing AP, (a_i's $\in$ Integer). If a_1 and $a_4 + a_3$ both are prime numbers and $a_5 = 31$, then

47. Number of such AP's possible having positive terms is/are
(A) 1
(B) 2
(C) 3
(D) 4

48. The greatest possible sum of 100 terms of AP is
(A) 34950
(B) 34951
(C) 34952
(D) 35950

Section – B (Maximum Marks: 12)
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. The length of two opposite edges of a tetrahedron ABCD are 3 4. If the angle between opposite edges is $\cos^{-1}\left(\frac{\sqrt{15}}{4}\right)$ and the volume of tetrahedron is 5 cubic units, then the shortest distance between the opposite edges of given length 3 and 4 is k units where k is

50. Let the vertices A, B, C of triangle ABC are represented by complex numbers p, q, r respectively in complex plane. If the angles at B and C are each equal to $\left(\frac{\pi-\alpha}{2}\right)$ and $(q-r)^2 = \left(2+\sqrt{3}\right)(p-q)(r-p)$, then $[2\alpha]$ is (Where [.] represents greatest integer function)

51. Let $K = \int_0^1 \frac{\left(x^{11}-x^6+x\right)}{\left(2x^{10}-3x^5+6\right)^{\frac{3}{5}}}dx$, then the value of $(12K)^5$ is

Section – C (Maximum Marks: 12)
*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Consider a triangle ABC having incentre I and excentre opposite to vertex A as I_1. If $II_1 = 20\sqrt{2}$, circum-radius of ΔABC is 10 and length of tangent from vertex A to the excircle opposite to vertex A is 24. Also sides opposite the angles satisfy b < c < a, then

52. The inradius of the triangle ABC is

53. Square of the length of median of triangle ABC through vertex C is

Question Stem for Question Nos. 54 and 55

Question Stem

Let S be the set containing all matrices $A = \left[a_{ij}\right]_{3\times 3}$ having entries from the set {0, 1, 2, 3, 4, 5} such that sum of all principle diagonal entries of $A \cdot A^T = 26$. Also B be the number of matrices in S having four distinct elements in AP. Given that $\alpha\beta = B$; ($\alpha, \beta \in N$)

54. Number of matrices in S are

55. Number of distinct ordered pairs (α, β) such that $\frac{\alpha}{\beta}$ is an integer are

Question Stem for Question Nos. 56 and 57

Question Stem

Let f(x) be a real valued differentiable function on R such that $e^{-x}f(x) = 109e^{-4} + 140e^{-x}\int_4^x \sqrt{t^2 + 4t + 17}\,dt$ $\forall$ x $\in$ R. If g(x) = $f^{-1}(x)$; ($f^{-1}(x)$ represents inverse of f(x)), then

56. The value of $\frac{1}{\sqrt{g'(109)}}$ is

57. The value of $(1089)^3\,|g''(109)|$ is